

TASK 3 REPORT

Model Validation, Overfitting Control & Hyperparameter Tuning

Dataset: California Housing dataset **Task focus:** Overfitting analysis, cross-validation, hyperparameter tuning, and model selection

1. Objective and Methodology

The objective of this task was to evaluate model generalization, identify signs of overfitting, and improve a Decision Tree Regressor through hyperparameter tuning. The California Housing dataset was used to predict the target variable, *HousePrice*, from the available housing features.

The dataset was divided into training and test subsets using an 80:20 split with `random_state=42`. Feature scaling was fitted on the training data and then applied to the test data. For cross-validation and grid search, a scikit-learn Pipeline combined `StandardScaler` with the Decision Tree Regressor, ensuring that scaling was fitted within each training fold rather than using information from its validation fold.

Workflow

- Load the California Housing dataset and separate features from the target.
- Split the original features into training and test sets.
- Fit `StandardScaler` on training features only; transform the training and test features.
- Fit a default Decision Tree and compare training and test RMSE.
- Estimate generalization with five-fold cross-validation.
- Use `GridSearchCV` to tune tree depth and minimum samples required for a split.
- Evaluate the selected model on the held-out test set and compare it with Linear Regression and Ridge Regression.

2. Overfitting Analysis

Overfitting occurs when a model learns the training examples too closely, including patterns that do not generalize to unseen data. A common warning sign is a very low training error combined with a noticeably higher test error.

Metric	Default Decision Tree result	Interpretation
Training RMSE	Approximately 0.000	The tree nearly memorized the training set.
Test RMSE	0.703	Prediction error was higher on unseen data.
5-fold CV RMSE	0.724	Provides a more robust estimate across validation folds.

The default tree achieved a training RMSE effectively equal to zero, but its test RMSE was about 0.703. This large train–test gap indicates overfitting: the unrestricted tree fit the training examples extremely closely but made less accurate predictions on unseen examples. Its five-fold cross-validation RMSE of approximately 0.724 also supports the need to control tree complexity.

Why the distinction matters: training performance alone is not sufficient for model selection. Validation and test performance provide evidence about how well the model is likely to generalize.

3. Hyperparameter Tuning Approach

GridSearchCV was used to search combinations of two Decision Tree hyperparameters. The search used five-fold cross-validation and negative root mean squared error as its scoring metric. Since scikit-learn reports this loss metric as a negative score, the reported RMSE is obtained by changing the sign.

Hyperparameter	Values searched	Purpose
max_depth	3, 5, 7, 10	Limits the maximum depth of the tree and helps control complexity.
min_samples_split	2, 5, 10	Sets the minimum number of samples required to split an internal node.

The grid contained 12 parameter combinations (4 depth values × 3 split thresholds). Each combination was assessed over five folds, and the configuration with the lowest cross-validation RMSE was selected.

Best configuration returned by GridSearchCV

max_depth = 10 min_samples_split = 10

The best cross-validation RMSE was **0.6375**. The tuned estimator was then evaluated on the held-out test set, which was not used to select the hyperparameters. This separation helps provide a less biased estimate of final model performance.

4. Final Model Comparison

Model	Test RMSE	Test R ²
Tuned Decision Tree	0.645	0.682
Ridge Regression	0.746	0.576
Linear Regression	0.746	0.576

RMSE measures prediction error in the target's units, with lower values indicating better performance. R² measures the proportion of target variance explained by the model on the test set; higher values are generally preferable. The tuned tree achieved the lowest test RMSE (0.645) and highest test R² (0.682) among the three evaluated models.

5. Final Model Justification

The Tuned Decision Tree Regressor was selected because it delivered the strongest test-set results in this comparison. Its RMSE of 0.645 was lower than the 0.746 reported for both Linear Regression and Ridge Regression, while its R^2 of 0.682 exceeded the 0.576 achieved by the two linear models.

The tuned model also has a more controlled structure than the default tree. Limiting maximum depth and requiring more samples for a split restricts the tree from creating overly specific branches. The selected configuration produced a best cross-validation RMSE of 0.6375 and a held-out test RMSE of 0.645, which are close in magnitude and provide encouraging evidence of generalization on this split.

Conclusion: For this experiment, the Tuned Decision Tree Regressor is the preferred model based on the lowest test RMSE and highest test R^2 among the evaluated candidates. The results demonstrate why training error alone is insufficient and how cross-validation and hyperparameter tuning can help control overfitting and improve predictive performance.

6. Limitations and Possible Improvements

- The results are based on one train-test split; repeating the experiment with different random splits can help assess stability.
- Only selected values of `max_depth` and `min_samples_split` were searched. A broader search could test additional configurations.
- Other useful controls, such as `min_samples_leaf` and `max_leaf_nodes`, could be explored.
- Other metrics such as MAE could provide complementary information alongside RMSE and R^2 .

Key Results at a Glance

Default tree test RMSE	Best CV RMSE	Tuned tree test RMSE	Tuned tree test R^2
0.703	0.6375	0.645	0.682